

Chapter 6: Anatomy of Flowering Plants

Dicot vs Monocot

BAIO LEARNING SYSTEM · Narendra Sir — Founder/CEO

Class Notes · Topic 8/8

Root: mono/oligoarch small pith (dicot) vs polyarch large pith (monocot)

Stem: ring VBs + pith (dicot) vs scattered VBs, no pith (monocot)

Leaf: dorsiventral, differentiated mesophyll (dicot) vs isobilateral, undifferentiated (monocot)

Golden rule: Open+Radial = dicot root; Open+Conjoint = dicot stem; Closed+Conjoint = monocot stem/leaf

Why Compare Dicots & Monocots?

It is convenient to study the transverse sections (T.S.) of the mature zones of dicotyledonous and monocotyledonous plants for a better understanding of how tissues are organised in roots, stems and leaves — and board exams frequently ask for these differences directly.

Root: Dicot vs Monocot

Both show epiblema, cortex and a barrel-shaped endodermis with Casparian strips. They differ in the **pericycle** (thick-walled in dicot, initiating cambium for secondary growth vs. secondary growth absent in monocot), the **pith** (small/inconspicuous in dicot vs large/well-developed in monocot), and the **vascular bundle** (monoarch/oligoarch in dicot vs polyarch, >6 patches, in monocot).

Stem: Dicot vs Monocot

The **hypodermis** is collenchymatous in dicot stems but sclerenchymatous in monocot stems. **Vascular bundles** are arranged in a ring in dicot stems (limited number, cambium present, open) but scattered throughout the ground tissue in monocot stems (numerous, cambium absent, closed). **Pith** is present in dicot stems but absent in monocot stems, where phloem parenchyma is also absent and water cavities are seen instead.

Leaf: Dicot vs Monocot

A dicot leaf is **dorsiventral**, with mesophyll differentiated into palisade and spongy parenchyma, more stomata on the abaxial surface, and reticulate venation. A monocot leaf is **isobilateral**, with undifferentiated mesophyll, roughly equal stomata on both surfaces, parallel venation, and characteristic **bulliform cells** that help the leaf roll up to conserve water.

Dicot vs Monocot – Master Comparison

Organ / Feature	Dicot	Monocot
Root vascular bundle	Monoarch–oligoarch (2–4), small pith	Polyarch (>6), large pith

Stem hypodermis	Collenchyma	Sclerenchyma
Stem vascular bundles	Ring, open, cambium present	Scattered, closed, cambium absent
Stem pith	Present	Absent
Leaf type	Dorsiventral	Isobilateral
Leaf mesophyll	Palisade + spongy (differentiated)	Undifferentiated
Venation	Reticulate	Parallel

3 Golden Rules

- Open vascular bundle (cambium present) → dicot
- Closed vascular bundle (no cambium) → monocot
- Radial VB → root; Conjoint VB → stem & leaf